



MICROSOFT INNOVATION CLUB, VIT CHENNAI

Hyperlocal Climate Risk-to-Action and Resilient Response Network

Sustainability & Climate

1. BACKGROUND

Floods, heatwaves, cyclones, landslides, glacial lake failures, marine heat events, and other climate hazards increasingly threaten communities. Weather forecasts and disaster bulletins may warn an entire city or district, but residents still don't know whether their street, village, campus, or travel route is at risk.

Responders struggle to identify which locations, roads, shelters, infrastructure, and populations need attention first. The necessary signals exist across weather feeds, satellite imagery, terrain maps, historical incidents, ocean observations, IoT sensors, emergency reports, and citizen observations — but arrive at *different times and scales, with varying reliability*, and during a severe event, cellular/power/internet access may fail.

The Problem

- Warnings are too broad — district-level forecasts don't explain risk to a specific neighborhood, route, or vulnerable population
- Data is fragmented and uncertain — environmental, infrastructure, and field information is hard to validate and combine in real time
- Monitoring is separated from action — maps may show danger without ranking rescues, routes, shelters, pumps, medical support, or evacuation decisions
- Conditions change faster than response plans — new rainfall, road closures, heat exposure, or community reports can make an earlier recommendation unsafe

- Communication fails at the last mile — app-only systems become unusable when people lack smartphones, data access, electricity, or functioning telecom infrastructure

Target beneficiaries: residents, commuters, disaster authorities, municipal teams, rescue services, healthcare workers, and NGOs, with infrastructure operators, researchers, insurers, and climate-adaptation planners as secondary beneficiaries.

OFFICIAL DOCUMENT

Prepared and issued for internal circulation and problem-statement reference by

MICROSOFT INNOVATION CLUB, VIT CHENNAI

Sustainability & Climate — Problem Statement

